

PRE PSPM

PHYSICS

DP024

2 hours

SET 6

LIST OF SELECTED FORMULAE

1. $a = -\omega^2 x$
2. $x = A \sin \omega t$
3. $v = \frac{dx}{dt}$
4. $a = \frac{dv}{dt} = \frac{d^2 x}{dt^2} = -\omega^2 x$
5. $\omega = 2\pi f = \frac{2\pi}{T}$
6. $k = \frac{2\pi}{\lambda}$
7. $v = \lambda f$
8. $y(x, t) = A \sin(\omega t \pm kx)$
9. $I = \frac{dQ}{dt}$
10. $\rho = \frac{RA}{l}$
11. $V = IR$
12. $R_{eff} = R_1 + R_2 + R_3 + \cdots + R_n$
13. $\frac{1}{R_{eff}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots + \frac{1}{R_n}$
14. $C = \frac{Q}{V}$
15. $\frac{1}{C_{eff}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots + \frac{1}{C_n}$
16. $C_{eff} = C_1 + C_2 + C_3 + \cdots + C_n$
17. $\tau = RC$
18. $Q = Q_o e^{-\frac{t}{RC}}$
19. $Q = Q_o \left(1 - e^{-\frac{t}{RC}}\right)$
20. $B = \frac{\mu_o I}{2\pi r}$
21. $B = \frac{\mu_o NI}{2R}$
22. $B = \frac{\mu_o NI}{L}$
23. $\phi = \vec{B} \cdot \vec{A}$
24. $\Phi = N\phi$
35. $\varepsilon = \frac{d\phi}{dt}$
26. $\varepsilon = Blv \sin \theta$
27. $\varepsilon = -NA \frac{dB}{dt}$
28. $\varepsilon = -NB \frac{dA}{dt}$
29. $R = 2f$
30. $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$
31. $m = \frac{h_i}{h_o} = -\frac{v}{u}$
32. $y_m = \frac{m\lambda D}{d}$
33. $y_m = \frac{\left(m + \frac{1}{2}\right)\lambda D}{d}$
34. $\Delta y = \frac{\lambda D}{d}$

LIST OF SELECTED CONSTANT VALUES

Speed of light in vacuum	c	=	$3.00 \times 10^8 \text{ m s}^{-1}$
Permeability constant	μ_o	=	$4\pi \times 10^{-7} \text{ H m}^{-1}$
Permittivity constant	ϵ_o	=	$8.85 \times 10^{-12} \text{ F m}^{-1}$
Elementary charge	e	=	$1.6 \times 10^{-19} \text{ C}$
Planck constant	h	=	$6.63 \times 10^{-34} \text{ J s}$
Electron mass	m_e	=	$9.11 \times 10^{-31} \text{ kg}$
Neutron mass	m_n	=	$1.672 \times 10^{-27} \text{ kg}$
Proton mass	m_p	=	$1.674 \times 10^{-27} \text{ kg}$
Deuteron mass	m_d	=	$3.34 \times 10^{-27} \text{ kg}$
Universal gas constant	R	=	$8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Rydberg constant	R_H	=	109678 cm^{-1}
Avogadro constant	N_A	=	$6.02 \times 10^{23} \text{ mol}$
Boltzmann constant	k	=	$1.38 \times 10^{-23} \text{ J K}^{-1}$
Gravitational constant	G	=	$6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Free-fall acceleration	g	=	9.81 m s^{-2}
Atomic mass unit	$1u$	=	$1.66 \times 10^{-27} \text{ kg}$
Electron volt	eV	=	$1.6 \times 10^{-19} \text{ J}$
Constant of proportionality for Coulomb's law	$k = \frac{1}{4\pi\epsilon_o}$	=	$9.0 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
Atmospheric pressure	atm	=	$1.013 \times 10^5 \text{ Pa}$
Density of water	ρ	=	1000 kg m^{-3}

INSTRUCTION: ANSWER ALL QUESTIONS

1. (a) The progressive wave with an amplitude of 4 cm, period of 0.1 s and wavelength of 0.12 m is moving to the right. Calculate
- the frequency
 - the angular frequency
 - the wave number
 - Write the wave equation.

[4 marks]

- (b) Two identical waves are travelling towards each other;

$$y_1 = 8 \sin \left(\frac{3}{5} \pi t - \frac{1}{3} \pi x \right)$$

$$y_2 = 8 \sin \left(\frac{3}{5} \pi t + \frac{1}{3} \pi x \right)$$

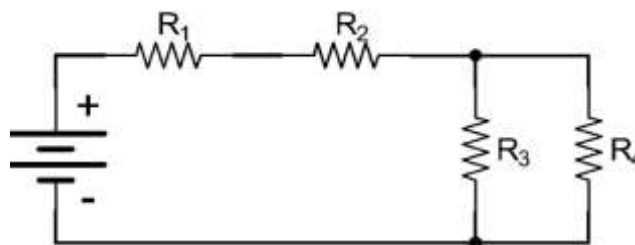
- Name the type of produced wave.
- Write the equation of the produced wave.
- Calculate the amplitude of the particle at $x = 2.5$ m.
- Calculate the wavelength of the wave.
- Calculate the speed of the wave.

[12 marks]

2. (a) A copper wire has a radius of 0.2 mm and a length of 8 m. Given the resistivity of copper is $1.72 \times 10^{-8} \Omega \text{ m}$.
- Calculate the resistance of the copper wire.
 - If a charge of 120 C passes through the wire within 1 minute, calculate the current that flows through the wire during that time.

[4 marks]

- (b) Four resistors are connected to a battery as shown in **FIGURE 1**. The resistance of R_1 and R_2 is 2Ω while R_3 and R_4 is 4Ω . The voltage supply is 12 V.

**FIGURE 1**

Calculate the

- effective resistance of the circuit
- total current in the circuit
- voltage across resistor R_1

[8 marks]

3.

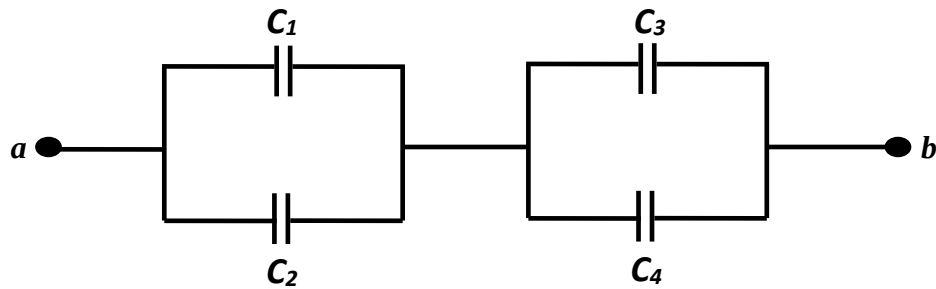


FIGURE 2

- (a) **FIGURE 2** shows four identical capacitors of $5\ \mu\text{F}$ connected between points a and b . Total charge in the circuit is $6 \times 10^{-5}\ \text{C}$.

Calculate the

- effective capacitance.
- voltage supply across points a and b .

[4 marks]

- (b) **FIGURE 3** shows a circuit of capacitor and resistor. The capacitor is fully charged. When the switch is closed, the capacitor is discharging.

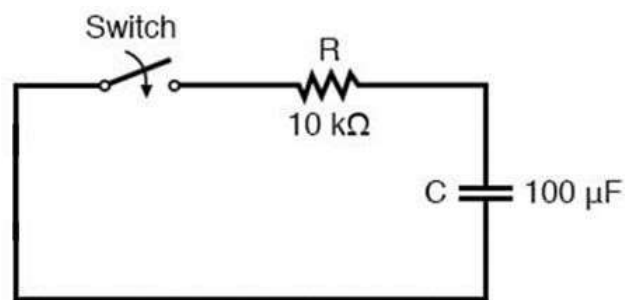
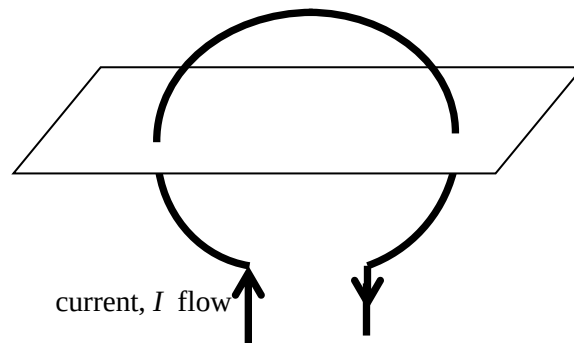


FIGURE 3

- Determine the time constant
- Calculate the time taken for the charge reduced by 75% of its maximum value
- Sketch a graph of charge on the capacitor against time.

[6 marks]

4. (a)

**FIGURE 4**

A student is provided with 2.0 m long wire. Then, the wire is bent into a circular coil of one turn as shown in **FIGURE 4**. A current of 0.15 A is flowing through it.

- (i) Sketch the magnetic field lines produced by circular loop.
- (ii) Calculate the radius of the circular coil.
- (iii) Calculate the strength of the magnetic field at the centre of the coil.

[7 marks]

(b)

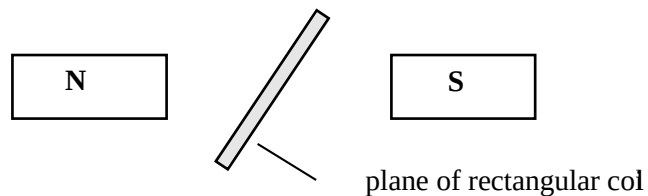
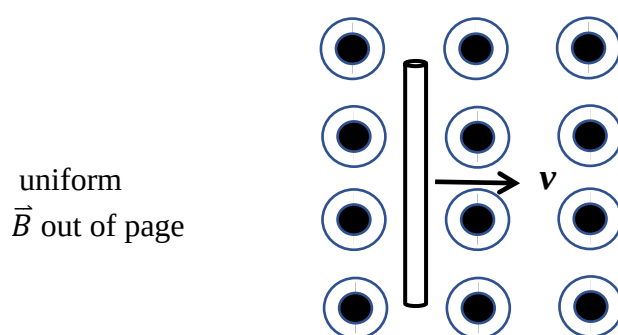
**FIGURE 5**

FIGURE 5 shows the plan view of a rectangular coil in a magnetic field. The area of the rectangular coil is 10 cm^2 and the magnetic field strength is 2.5 T. Calculate the magnetic flux through the coil if the plane makes 50° angle to the magnetic field.

[2 marks]

(c)

**FIGURE 6**

A 30 cm long metal rod AB is moved at speed of 50 m s^{-1} across a uniform magnetic field strength of 0.25 T. The motion of the rod is perpendicular to the magnetic field as shown in **FIGURE 6**.

- (i) Calculate the motional induced emf in the rod.
- (ii) If the rod is connected in series to the resistor of resistance 10Ω , calculate the induced current.
- (iii) State the direction of the induced current.

[5 marks]

5. (a) A pencil of height 12 cm is placed at the centre of curvature of a concave mirror with focal length 30 cm.

- (i) Calculate the position of the image.
- (ii) Calculate the image size.
- (iii) State two (2) characteristics of the image.
- (iv) Sketch and label a ray diagram to show the formation of the image.

[8 marks]

- (b) An inverted image is produced at a distance of 40 cm from an object by a convex lens. If the magnification is 4, calculate the

- (i) object distance.
- (ii) image distance.
- (iii) focal length of the lens.

[6 marks]

6. (a)

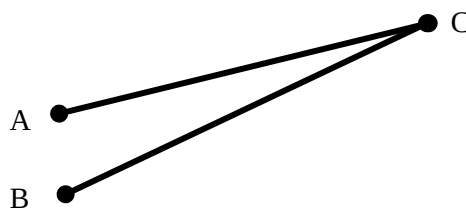
**FIGURE 7**

FIGURE 7 shows two paths of coherent lights from points A and B that produce an interference pattern at point C. Determine whether it is a constructive or destructive interference if AC and BC are 2.2λ and 5.7λ respectively.

[2 marks]

(b)

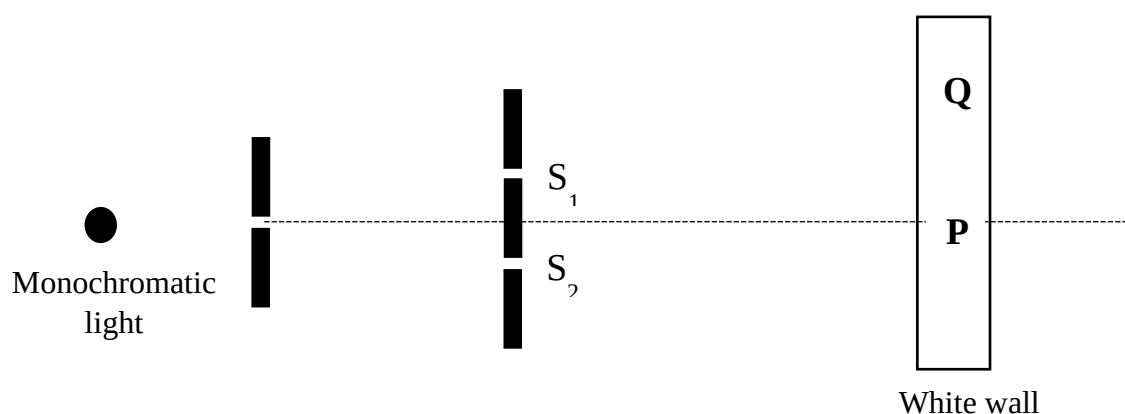
**FIGURE 8**

FIGURE 8 shows a monochromatic light passes through two narrow slits, S_1 and S_2 that are 0.1 mm apart. The interference pattern is observed on a white wall 1.2 m from the slits. A bright fringe is located at point P and a dark fringe is observed at point Q. The separation between two adjacent bright interference fringes is 6.0 mm .

- (i) Calculate the wavelength of the light.
- (ii) Calculate the distance of the fourth bright fringe from point P.
- (iii) Calculate the separation between the zeroth-order minimum and first order maximum.
- (iv) When a light source of wavelength 560 nm is used, the 40^{th} dark fringe is observed at Q. However, when a light source of wavelength, λ_1 is used, the 35^{th} dark fringe is observed at Q. Calculate the value of λ_2 .
- (v) What is the change that can be observed if the distance between coherent sources is reduced?

[12 marks]

End Of Question